

Finite Resonance Certificates and Sharp Analytic Normalization for Hahn–Fuchsian Systems

Accumulating exponents, unavoidable logarithms,
and a summability-threshold crossover

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29 September 2026

Abstract

We study linear systems $xY' = (A_0 + A_+(x))Y$ whose positive real exponents form a well-ordered set, without requiring local finiteness. For real-spectrum A_0 , we construct a normalized Hahn-series gauge and a finite resonant normal form. A single nilpotent matrix determines the complete filtration by logarithmic degree. Its entries depend on only finitely many original coefficients, even when infinitely many exponents lie below the largest resonance.

The central analytic result is a sharp universal criterion. Every absolutely convergent, semigroup-supported coefficient series has an absolutely convergent normalized gauge if and only if the nonresonant semigroup exponents are uniformly separated from the eigenvalue differences. For a well-ordered real semigroup, failure means accumulation from below at a positive eigenvalue difference. A rank-one matrix perturbation gives arbitrarily small counterexamples whenever this condition fails.

An explicit two-dimensional family has coefficient weights n^{-p} and exponents $2 - 1/n$. Its formal normalizer converges exactly when $p > 2$, although a renormalized analytic solution exists for every $p > 1$ and realizes all finite asymptotic prefixes. We derive its truncation crossover on the scale $N \asymp \log(1/x)$, including a uniform Euler–Maclaurin error bound. All general claims are proved conventionally. Reproducible finite symbolic checks and numerical illustrations accompany the article. Classical Fuchsian normal forms and the repository’s earlier finite-obstruction work are explicitly acknowledged; an exhaustive claim of historical priority is not made.

Scope. Positive real Hahn exponents; a constant matrix with real spectrum; one logarithmic indeterminate. This is not a solution of arbitrary transseries differential equations and is not a Lean formalization.

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1 The research question and its place in ProveIt

A formal transseries calculation and an analytic solution are different objects. A well-ordered support makes coefficientwise algebra meaningful; it does not bound the coefficients produced by division. This distinction becomes decisive when exponents accumulate at a finite value. The familiar Frobenius recurrence then meets a phenomenon absent from ordinary Taylor series: infinitely many different exponents can approach one fixed resonance.

The relevant ProveIt materials already develop asymptotic scales, flatness, polynomial–logarithmic block calculus, Hahn-support lemmas, and inversion methods. Their status documentation distinguishes actual Lean results from human proofs and unformalized extensions [1, 2, 3]. A recently filed companion, *Finite Residue Obstructions and Logarithmic-Depth Promotion in Hahn Transseries*, treats scalar differential operators by a finite obstruction matrix under a uniform outer-exponent decrease [4]. Consequently, the mere assertion that resonances can have finite obstruction data would not be a substantive new contribution to this repository.

Here the principal question is instead the following:

When does a support-controlled formal Fuchsian normalization automatically preserve absolute convergence, and what exactly goes wrong when it does not?

Two additional questions make the answer useful. Can all unavoidable logarithms be certified from finite input data even for non-locally-finite supports? Can a divergent normalizer nevertheless describe the finite asymptotic prefixes of a genuine analytic solution, with an explicit truncation law near an accumulation exponent?

These are concrete extension questions formulated in this article. They are not attributed to the repository as verbatim named open problems, and we do not claim to settle a famous general conjecture in transseries theory. Theorems 6.2, 5.1, and 9.1 answer them in the class specified below.

1.1 What is classical and what is proposed here

For ordinary integer exponents, reduction of a Fuchsian system to a resonant polynomial normal form is classical. The entrywise resonance rule and a product of two constant-matrix powers are recorded, for example, in Tanny and Yakovenko’s discussion of local normal forms [5, p. 6]. Our construction has that theory as a special case; neither the Frobenius method nor its integer-exponent normal form is claimed as a discovery.

General fixed-point methods for generalized series are also established; van der Hoeven proves an implicit-function theorem for Noetherian operators [6]. We give the particular support and valuation arguments needed here rather than claiming a new general Hahn fixed-point theorem. Convergence for Hahn series is itself an established subject, studied in other settings by Joswig and Smith [7].

The proposed contribution is the combination of an exact semigroup–spectrum criterion for *universal absolute convergence*, its arbitrarily small counterexamples, a finite-ancestry refinement of the resonance certificate, and the explicit crossover analysis of a normalizer at an accumulation threshold. Full proofs are supplied; historical novelty beyond the sources inspected remains to be independently assessed. In particular, the article should not be cited as an independently peer-reviewed breakthrough.

1.2 Results in one view

Write $D = x d/dx$ and decompose the constant matrix as $A_0 = S + N$, where S is diagonal with real entries and N is nilpotent and commutes with S . The formal construction produces

$$Y = H(x)x^S \exp(B \log x), \quad B^m = 0. \quad (1.1)$$

The coefficient support of $H - I$ stays in the additive semigroup generated by that of A_+ . The complete logarithmic filtration is

$$\dim_{\mathbb{C}}\{\text{solutions of logarithmic degree at most } d\} = \dim_{\mathbb{C}} \ker B^{d+1}. \quad (1.2)$$

Moreover, B is a polynomial in finitely many input coefficients once the constant matrix and exponent data are fixed.

For a fixed positive well-ordered semigroup Γ , the analytic criterion is

$$\inf_{\substack{\gamma \in \Gamma_{>0}, \\ 1 \leq i, j \leq m \\ \gamma \neq \lambda_i - \lambda_j}} |\gamma - (\lambda_i - \lambda_j)| > 0. \quad (1.3)$$

There are two essential qualifications. This is a universal assertion over all coefficient series supported in $\Gamma_{>0}$, not just a specified smaller generating set. It concerns the normalized gauge constructed here and absolute generalized-series convergence, not the existence of arbitrary analytic solutions of the differential equation.

2 Hahn supports and the differential model

2.1 Orientation and allowed supports

Throughout, x is a positive infinitesimal; analytically it tends to zero through positive values or lies on a chosen logarithmic cover of a punctured disk. A Hahn series is written

$$f = \sum_{\alpha \in \mathbb{R}} f_{\alpha} x^{\alpha}$$

with well-ordered support in the usual order on $\mathbb{R}$. Thus smaller exponents are dominant. This is the reverse of the common large-variable monomial convention, and converting between the two reverses the support order.

Let $E \subset \mathbb{R}_{>0}$ be nonempty and well ordered, and set

$$\delta = \min E > 0, \quad \Gamma = \langle E \rangle = \{0\} \cup \bigcup_{k \geq 1} \{\alpha_1 + \cdots + \alpha_k : \alpha_i \in E\}. \quad (2.1)$$

Neither E nor Γ is assumed locally finite. For instance, $E = \{2 - 1/n : n \geq 1\}$ is well ordered but has infinitely many members below 2. This type of support is indispensable for the sharpness result.

Lemma 2.1 (Support and finite words). *The semigroup Γ is well ordered. For each $\gamma \in \Gamma$, there are only finitely many ordered words $(\alpha_1, \dots, \alpha_k)$ in E whose sum is γ . Every such word has length at most $\lfloor \gamma/\delta \rfloor$.*

Proof. The two-set form of Neumann’s lemma says that the sum of two well-ordered subsets of an ordered abelian group is well ordered and that any fixed element has only finitely many decompositions with one term in each set. The repository’s additive support lemmas provide

precisely this two-factor bridge [2]. Induction therefore proves both assertions for each fixed word length k .

Positivity gives $k\delta \leq \gamma$. Hence only finitely many lengths can represent a fixed γ , proving the finite-word assertion. To prove well ordering of Γ , let $U \subset \Gamma$ be nonempty and choose $u \in U$. All elements of $U \cap [0, u]$ belong to the finite union $\bigcup_{0 \leq k \leq \lfloor u/\delta \rfloor} kE$, which is well ordered. The least element of this intersection is also the least element of U . $\square$

The proof uses the Archimedean order of the real exponents in an essential way: the common positive lower bound controls word length under a finite cutoff. The analogous assertion with an infinitesimal lower bound in a higher-rank ordered group requires a different hypothesis.

2.2 The ambient differential algebra

Let

$$\mathcal{K} = \mathbb{C}((x^{\mathbb{R}}))$$

be the Hahn field, and let $\mathcal{K}[L]$ be the polynomial ring in an algebraically independent variable L . Define the strong derivation

$$D\left(\sum f_{\alpha}x^{\alpha}\right) = \sum \alpha f_{\alpha}x^{\alpha}, \quad DL = 1. \quad (2.2)$$

The multiplication rule is meaningful because the coefficient convolutions are finite. The Leibniz rule follows by the equality $\alpha + \beta = \gamma$ in each such convolution. Analytically, L will be $\log x$.

Let $\mathcal{H}_{\Gamma}$ denote the Hahn-series subring with support in Γ , and let $\mathfrak{m}_{\Gamma}$ be its positive-support ideal. For a nonzero series or matrix series, $v(f)$ is the least exponent with nonzero coefficient, with $v(0) = +\infty$. Then

$$v(f + g) \geq \min(v(f), v(g)), \quad v(fg) \geq v(f) + v(g).$$

Matrices need the inequality, rather than equality, because leading matrix products can vanish.

Lemma 2.2 (Positive-support limits and inverses). *Suppose F_n are matrix series supported in Γ and $v(F_{n+1} - F_n) \rightarrow +\infty$. Each coefficient eventually stabilizes, and the coefficientwise limit is a Hahn series supported in Γ . If $h \in \text{Mat}_m(\mathfrak{m}_{\Gamma})$, then*

$$(I + h)^{-1} = \sum_{k \geq 0} (-h)^k$$

is a well-defined matrix Hahn series supported in Γ .

Proof. Any fixed exponent lies below the valuations of all sufficiently late differences. The limit support is a subset of the well-ordered set Γ . For the inverse, $v(h^k) \geq k\delta$, so the displayed sum is coefficientwise finite. Multiplication by $I + h$ telescopes, and the remaining power tends coefficientwise to zero. $\square$

Lemma 2.3 (Constants and the missing Euler residue). *The constant field of $\mathcal{K}$ under D is $\mathbb{C}$, and no derivative in $\mathcal{K}$ has a nonzero coefficient at x^0 . The constants of $\mathcal{K}[L]$ are also exactly $\mathbb{C}$.*

Proof. The first two statements follow immediately from $[x^{\alpha}]Df = \alpha f_{\alpha}$. For the third, suppose $g = \sum_{j=0}^d g_j L^j$ is nonzero and $Dg = 0$. Its highest coefficient satisfies $Dg_d = 0$, hence $g_d \in \mathbb{C}$. If $d > 0$, the coefficient of L^{d-1} gives $Dg_{d-1} = -dg_d$, impossible because the right side has a nonzero x^0 coefficient. Thus $d = 0$ and $g \in \mathbb{C}$. $\square$

This elementary residue obstruction is the scalar mechanism behind the logarithmic variable. The matrix certificate below records how different spectral levels couple that mechanism.

Editorial note (ProveIt, 2026-09-29). Lemma 2.3 is the depth-0 case, written for the Euler derivation $D = x d/dx$, of results in the canonical volume (Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex), which this article could not retrieve: since $[x^0](x dF/dx) = [x^{-1}] dF/dx$, its residue statement is equation plt:eq:ext-residue of Theorem plt:thm:ext-tower-strict at $r = 0$, and its statements on constants are Corollary plt:cor:dif-kernel. At depth $n \geq 1$ the residue-obstruction package Analysis/Transseries/docs/series-and-transseries/ResidueObstructions_Logarithmic_Depth_Promotion/ (residue_fredholm.tex, Theorem thm:residue) sharpens the same residue to an exact sequence. The volume uses the large-variable convention; the Euler-form statement is unchanged by $x \mapsto 1/x$, which only changes the sign of D .

3 The homological equation and normalized gauge

3.1 The constant matrix and its spectral blocks

Consider

$$DY = (A_0 + A_+(x))Y, \quad A_+(x) = \sum_{\alpha \in E} A_\alpha x^\alpha. \quad (3.1)$$

Zero coefficients are allowed. Assume $A_0 \in \text{Mat}_m(\mathbb{C})$ has only real eigenvalues. Choose once and for all a Jordan basis and write

$$A_0 = S + N, \quad S = \text{diag}(\lambda_1, \dots, \lambda_m), \quad [S, N] = 0, \quad N^m = 0. \quad (3.2)$$

Changing this constant basis changes the normalization convention; all uniqueness assertions below refer to the chosen basis and projections.

For $\gamma > 0$, let P_γ be the coordinate projection

$$(P_\gamma M)_{ij} = \begin{cases} M_{ij}, & \lambda_i - \lambda_j = \gamma, \\ 0, & \text{otherwise,} \end{cases} \quad Q_\gamma = I - P_\gamma. \quad (3.3)$$

Here the I in Q_γ is the identity operator on matrix space. Define the finite resonance set

$$\mathcal{R} = \{\lambda_i - \lambda_j > 0 : 1 \leq i, j \leq m\} \cap \Gamma, \quad r_* = \max(\mathcal{R} \cup \{0\}). \quad (3.4)$$

The homological operator is

$$\mathcal{L}_\gamma = \gamma I - \text{ad}(A_0), \quad \text{ad}(A_0)M = A_0 M - M A_0.$$

Both $\text{ad}(N)$ and $\mathcal{L}_\gamma$ preserve every eigenspace of $\text{ad}(S)$ because $[S, N] = 0$.

Lemma 3.1 (Explicit nonresonant inverse). *Define the operator $\mathcal{D}_\gamma$ by*

$$(\mathcal{D}_\gamma M)_{ij} = \begin{cases} \frac{M_{ij}}{\gamma - (\lambda_i - \lambda_j)}, & \gamma \neq \lambda_i - \lambda_j, \\ 0, & \gamma = \lambda_i - \lambda_j. \end{cases}$$

Then

$$T_\gamma = \sum_{k=0}^{2m-2} \mathcal{D}_\gamma^{k+1} (\text{ad } N)^k \quad (3.5)$$

satisfies

$$\mathcal{L}_\gamma T_\gamma = Q_\gamma, \quad P_\gamma T_\gamma = 0.$$

For each F , the unique pair (H, R) satisfying $P_\gamma H = 0$, $R = P_\gamma R$, and $\mathcal{L}_\gamma H + R = F$ is $H = T_\gamma F$, $R = P_\gamma F$.

Proof. Left and right multiplication by N commute, so the binomial expansion of $(\text{ad } N)^{2m-1}$ has in every term either at least m left factors or at least m right factors. Thus $(\text{ad } N)^{2m-1} = 0$. On the range of Q_γ , $\gamma I - \text{ad}(S)$ is invertible with inverse $\mathcal{D}_\gamma$, which commutes with $\text{ad } N$. The finite geometric sum (3.5) is therefore the inverse of $(\gamma I - \text{ad } S) - \text{ad } N$ on that range.

The resonant projection of $\mathcal{L}_\gamma H$ vanishes when $P_\gamma H = 0$, since the operator preserves the spectral blocks. Projecting the equation gives $R = P_\gamma F$. The complementary equation has the stated unique solution. $\square$

The finite geometric sum parallels the repository's explicit inverse of $\partial_L - c$ on bounded-degree logarithmic polynomials [3]; here the nilpotent operator is $\text{ad } N$ instead of polynomial differentiation.

3.2 The formal normalization theorem

Theorem 3.2 (Normalized Hahn–Fuchsian form). *For (3.1)–(3.2), there is a unique pair*

$$H = I + \sum_{\gamma \in \Gamma_{>0}} H_\gamma x^\gamma, \quad R(x) = \sum_{\rho \in \mathcal{R}} R_\rho x^\rho \quad (3.6)$$

with $P_\gamma H_\gamma = 0$ and $P_\rho R_\rho = R_\rho$, such that

$$DH = (A_0 + A_+)H - H(A_0 + R). \quad (3.7)$$

All series are well-defined Hahn series. No exponent outside Γ is introduced into H or R .

Proof. Extend A_γ to be zero outside E . Equating the coefficient of x^γ in (3.7) gives

$$F_\gamma = A_\gamma + \sum_{\substack{\alpha+\beta=\gamma \\ \alpha, \beta > 0}} A_\alpha H_\beta - \sum_{\substack{\beta+\eta=\gamma \\ \beta, \eta > 0}} H_\beta R_\eta, \quad (3.8)$$

$$H_\gamma = T_\gamma F_\gamma, \quad R_\gamma = P_\gamma F_\gamma. \quad (3.9)$$

Every coefficient on the right of (3.8), except the input A_γ , has exponent strictly less than γ . The sums are finite by Lemma 2.1. Well-ordered recursion and Lemma 3.1 give existence and uniqueness. If $\gamma \notin \mathcal{R}$, then $P_\gamma = 0$, so $R_\gamma = 0$. The resulting supports are subsets of Γ , proving their admissibility.

For a second construction that will also give an analytic estimate, let $\mathcal{P}$ and $\mathcal{T}$ act coefficientwise by P_γ and T_γ on positive-support matrix series. Set

$$\Phi(F) = A_+ + A_+(\mathcal{T}F) - (\mathcal{T}F)(\mathcal{P}F), \quad F_0 = 0, \quad F_{n+1} = \Phi(F_n). \quad (3.10)$$

All F_n have support in $\Gamma_{>0}$. Both coefficientwise operators preserve or increase valuation. If F, G have positive support, then

$$\Phi(F) - \Phi(G) = A_+ \mathcal{T}(F - G) - \mathcal{T}(F - G) \mathcal{P}F - (\mathcal{T}G) \mathcal{P}(F - G).$$

Consequently,

$$v(\Phi(F) - \Phi(G)) \geq v(F - G) + \delta. \quad (3.11)$$

It follows by induction that $v(F_{n+1} - F_n) \geq (n+1)\delta$. Lemma 2.2 gives a limit F ; finite coefficient convolutions show $F = \Phi(F)$. Take $H = I + \mathcal{T}F$ and $R = \mathcal{P}F$. Lemma 3.1 yields (3.7). A difference of two fixed points would have its finite valuation increased by at least δ , a contradiction. This also proves uniqueness without invoking transfinite recursion separately. $\square$

Corollary 3.3 (Finite iteration count for the normal form). *The iterates in (3.10) satisfy*

$$v(F - F_n) \geq (n+1)\delta.$$

In particular, $R = \mathcal{P}F_n$ as soon as $(n+1)\delta > r_$.*

Proof. All differences in the tail beginning with $F_{n+1} - F_n$ have valuation at least $(n+1)\delta$. The projection $\mathcal{P}$ retains only exponents at most r_* . $\square$

Remark 3.4. A finite number of formal iterations does not mean that each whole iterate has finite support. When exponents accumulate, even its part below r_* can be infinite. The genuinely finite input certificate is proved in Section 5, not by silently replacing well ordering with local finiteness.

4 A finite matrix controls every logarithm

Theorem 4.1 (Fundamental matrix and logarithmic filtration). *For the normalized pair (H, R) , put*

$$B = N + \sum_{\rho \in \mathcal{R}} R_\rho. \quad (4.1)$$

Then $B^m = 0$, and

$$Y = H(x)x^S e^{BL}, \quad x^S = \text{diag}(x^{\lambda_1}, \dots, x^{\lambda_m}), \quad e^{BL} = \sum_{k=0}^{m-1} \frac{B^k L^k}{k!} \quad (4.2)$$

is a fundamental matrix over $\mathcal{K}[L]$. Every solution vector with entries in $\mathcal{K}[L]$ is uniquely Yc for $c \in \mathbb{C}^m$.

For each integer $d \geq 0$, the space of solutions of logarithmic degree at most d has dimension

$$\dim_{\mathbb{C}} \text{Sol}_{\leq d} = \dim_{\mathbb{C}} \ker B^{d+1}. \quad (4.3)$$

A logarithm-free fundamental matrix exists in this class if and only if $B = 0$. If $\nu(B)$ is the least positive integer with $B^{\nu(B)} = 0$ (with $\nu(0) = 1$), then every fundamental matrix in $\mathcal{K}[L]$ has maximum logarithmic degree exactly $\nu(B) - 1 \leq m - 1$.

Proof. The matrix N preserves each eigenspace of S , whereas R_ρ maps the λ eigenspace into the $\lambda + \rho$ eigenspace. Order the distinct eigenvalues increasingly. The matrix B is block triangular, with nilpotent diagonal blocks inherited from N . Its characteristic polynomial is z^m , so the Cayley–Hamilton theorem gives $B^m = 0$.

The spectral support rule gives

$$x^S B x^{-S} = N + \sum_{\rho \in \mathcal{R}} x^\rho R_\rho = N + R(x). \quad (4.4)$$

Differentiating $x^S e^{BL}$, using $DL = 1$, yields

$$D(x^S e^{BL}) = (S + N + R)x^S e^{BL} = (A_0 + R)x^S e^{BL}.$$

Combining this with (3.7) proves that Y solves (3.1). The inverse

$$Y^{-1} = e^{-BL}x^{-S}H^{-1}$$

has entries in $\mathcal{K}[L]$ by Lemma 2.2; hence Y is a fundamental matrix.

If y is any other solution in this algebra, then $D(Y^{-1}y) = 0$. Lemma 2.3 gives $Y^{-1}y = c \in \mathbb{C}^m$. The coefficient of L^k in Yc is $Hx^S B^k c/k!$. Because Hx^S is invertible and independent of L , this coefficient is zero exactly when $B^k c = 0$. Thus $\deg_L(Yc) \leq d$ is equivalent to $B^{d+1}c = 0$, which proves (4.3).

Any fundamental matrix is YC with $C \in \mathrm{GL}_m(\mathbb{C})$. Its top nonzero logarithmic coefficient is $Hx^S B^{\nu(B)-1}C/(\nu(B)-1)!$, which cannot vanish. The remaining assertions follow. $\square$

Corollary 4.2 (Invariance under log-free gauges). *Multiplication of a solution vector by an invertible matrix over $\mathcal{K}$ preserves its logarithmic degree. Hence the dimensions in (4.3) are invariants of log-free gauge equivalence, even though the entries of the normalized matrix B depend on the chosen normalization.*

Proof. An invertible log-free matrix cannot kill the highest nonzero coefficient of a polynomial in L . $\square$

Example 4.3 (The bound $m-1$ is attained). Let $S = \mathrm{diag}(0, 1, \dots, m-1)$, $N = 0$, and $J = \sum_{j=1}^{m-1} E_{j+1,j}$, where E_{ij} is a matrix unit. For $A_+ = xJ$, all entries are resonant. The normalized gauge is $H = I$, the normal form is $R = xJ$, and $B = J$. Since $J^{m-1} \neq 0$ and $J^m = 0$, logarithmic degree $m-1$ is unavoidable. The result is an exact identity, not a merely asymptotic model.

Remark 4.4 (The ambient algebra matters). The assertion is about polynomial dependence on the one indeterminate L over the log-free Hahn field $\mathcal{K}$. It does not identify L with an element already in $\mathcal{K}$, nor does it claim that every possible analytic representation must visibly display a logarithm. An unrestricted special function can hide one. The filtration is meaningful because the coefficient algebra and allowed log-free gauges have been fixed.

5 Finite ancestry despite infinitely many low exponents

Define the critical ancestry set

$$E_{\mathrm{crit}} = \bigcup_{\rho \in \mathcal{R}} \bigcup_{\substack{k \geq 1, \alpha_1, \dots, \alpha_k \in E \\ \alpha_1 + \dots + \alpha_k = \rho}} \{\alpha_1, \dots, \alpha_k\}. \quad (5.1)$$

The union over an empty resonance set is empty.

Theorem 5.1 (Finite-ancestry certificate). *The set E_{crit} is finite. For fixed S, N, E , each matrix R_ρ is an ordinary polynomial in the entries of $\{A_\alpha : \alpha \in E_{\mathrm{crit}}\}$, of degree at most $\lfloor \rho/\delta \rfloor$. Its scalar coefficients depend on the constant matrix and the finitely many intermediate exponent differences. In particular, all logarithmic obstruction data are independent of input coefficients outside E_{crit} .*

For a family obtained by varying these finitely many matrix entries,

$$\{\text{maximum logarithmic degree} \leq d\} = \{B^{d+1} = 0\} \quad (5.2)$$

is an algebraic locus. The ranks of the powers of B , and hence the entire logarithmic filtration, are determined by polynomial minors.

Proof. There are finitely many resonances. Lemma 2.1 gives finitely many words ending at each one, so (5.1) is finite.

Expand the recurrence (3.8)–(3.9) by repeated substitution. The operators T_γ and P_γ are linear and do not shift exponents. Every product term therefore has input leaves $A_{\alpha_1}, \dots, A_{\alpha_k}$ whose exponent sum is the output exponent. Each further substitution replaces a positive exponent by positive smaller exponents. The expansion ends after at most $\lfloor \gamma/\delta \rfloor$ input leaves. Finiteness of the possible words also gives finiteness of all partial exponent sums, so the expression at each fixed output exponent is an ordinary finite polynomial, not an infinite coefficient formula.

At output ρ , all leaves belong to E_{crit} by definition. The only divisions are the nonzero scalar denominators in T_γ , which depend on S and intermediate exponents rather than on input coefficient variables. Thus the polynomial degree is bounded by the number of leaves. Since $B = N + \sum R_\rho$, its entries are polynomials in finite data. Theorem 4.1 proves (5.2); the standard minor criteria for matrix rank prove the last assertion. $\square$

5.1 What the finite statement does not assert

For an arbitrary infinitely presented set of real exponents, it can be impossible to decide whether a proposed number is a finite sum of members. The theorem supplies finite mathematical dependence; it does not supply an unconditional terminating algorithm for discovering the critical set from an arbitrary oracle. The accompanying implementation restricts to finite rational input supports, where exact equality and bounded enumeration are decidable.

There is a useful distinction between three cutoffs. The largest resonance r_* bounds where normal-form output can occur. The number $\lfloor r_*/\delta \rfloor$ bounds the number of input factors in an obstruction. Neither statement says that all exponents below r_* must be enumerated. The critical ancestry set selects just the finitely many exponents that can actually reach a resonance by addition.

5.2 Relation to the earlier residue-matrix method

The repository’s residue-obstruction article already uses a positive exponent decrement to bound perturbative paths between finitely many real roots [4]. Our finite-ancestry theorem is aligned with that principle. The distinction is the present matrix normal form, the complete log-degree filtration via powers of B , and especially the analytic criterion in the next section. The previous scalar fixed-depth problem and the present universal absolute-convergence problem are not interchangeable.

Editorial note (ProveIt, 2026-09-29). This article knew the residue-obstruction package only through its README. Its source, `Analysis/Transseries/docs/series-and-transseries/Residue_Obstructions_Logarithmic_Depth_Promotion/residue_fredholm.tex`, poses in its Section 11 two questions that Theorem 4.1 answers for the present class. “The exact minimal power of the promoted logarithm” asks for residue matrices whose ranks recover the full filtration by degree in the new logarithm: here, for homogeneous solutions, it is $\dim \text{Sol}_{\leq d} = \dim \ker B^{d+1}$ (particular solutions of forced equations are not treated). “Matrix systems and nonreal indicial roots” asks which invariant controls the new-logarithm degree: here it is the nilpotency index $\nu(B)$, which combines Jordan chains in N with passages between root levels through the R_ρ (real spectrum only). The two articles are consistent but not nested: that article works at depth $n \geq 1$ with scalar operators $P(xD) + \sum a_j(xD)^j$, this one at depth 0 with first-order systems. At $d = 0$, $\dim \ker B = m - \text{rank } B$ is the analogue of its kernel

dimension $r - \text{rank } M$ (Theorem `thm:fredholm`); Example 4.3 is the analogue of its Jordan-shift family (Theorem `thm:sharp-matrix`); and when $N = 0$, B is strictly block-triangular over the s distinct eigenvalues, so the logarithmic degree is at most $s - 1$, its homogeneous bound (Theorem `thm:promotion`). With $N \neq 0$ the degree can reach $m - 1 > s - 1$, which is no contradiction: there the multiplicity logarithms already lie in the base field.

6 A sharp criterion for universal analytic normalization

6.1 The absolute generalized-series algebra

Use the induced column-sum matrix norm

$$\|M\|_1 = \max_j \sum_i |M_{ij}|,$$

so $\|I\|_1 = 1$, matrix multiplication is submultiplicative, and every coordinate mask P_γ or Q_γ is contractive. For $r > 0$ set

$$\|F\|_r = \sum_{\gamma \in \Gamma_{>0}} \|F_\gamma\|_1 r^\gamma. \quad (6.1)$$

A series is called absolutely convergent at radius r when this norm is finite. Together with a constant term, these series form a Banach algebra under convolution. A well-ordered subset of $\mathbb{R}$ is countable, so the notation is an ordinary countable absolute sum; equivalently it can be read as a supremum over finite subsums.

For real exponents and a chosen branch, $|x^\gamma| = |x|^\gamma$. Absolute convergence therefore gives holomorphic evaluation on the logarithmic cover of $0 < |x| < r$. The Euler derivative converges normally on every smaller radius $r' < r$, because $\sup_{\gamma \geq 0} \gamma(r'/r)^\gamma < \infty$. Products and the Euler derivative may then be evaluated termwise.

Define

$$\eta(\Gamma, S) = \inf \left\{ |\gamma - (\lambda_i - \lambda_j)| : \begin{array}{l} \gamma \in \Gamma_{>0}, \ 1 \leq i, j \leq m, \\ \gamma \neq \lambda_i - \lambda_j \end{array} \right\}. \quad (6.2)$$

The exclusion of exact resonances is crucial: exact resonances are put in R , not divided by.

Lemma 6.1 (Which accumulations matter). *The number $\eta(\Gamma, S)$ is positive if and only if no positive eigenvalue difference is a limit of a strictly increasing sequence of nonresonant exponents in Γ .*

Proof. Such an increasing sequence forces the infimum to be zero. Conversely, if the infimum is zero, finiteness of the set of eigenvalue differences provides one difference ρ and distinct exponents $\gamma_n \in \Gamma_{>0}$ tending to ρ , none equal to it. Since $\gamma_n \geq \delta$, the difference ρ is positive. A subsequence approaching ρ from above would contain an infinite strictly decreasing subsequence, contrary to well ordering of Γ . There are therefore infinitely many terms below ρ , from which one extracts a strictly increasing subsequence tending to ρ . $\square$

Theorem 6.2 (Universal convergence: necessary and sufficient). *Fix a nontrivial semigroup $\Gamma \subset \mathbb{R}_{\geq 0}$ generated as in (2.1), and fix $A_0 = S + N$ as in (3.2). The following statements are equivalent.*

- (i) $\eta(\Gamma, S) > 0$.
- (ii) *For every matrix series A_+ supported in $\Gamma_{>0}$ that is absolutely convergent at some radius, the normalized formal gauge H of Theorem 3.2 is absolutely convergent at some positive radius.*

If (i) fails, counterexamples in (ii) can have arbitrarily small norm at radius 1 and can be chosen proportional to one square-zero rank-one constant matrix. Their coefficient series are absolutely convergent at every positive radius.

6.2 Proof of sufficiency with quantitative bounds

Assume $\eta > 0$ and put

$$C = \sum_{k=0}^{2m-2} \frac{(2\|N\|_1)^k}{\eta^{k+1}}. \quad (6.3)$$

The powers are ordinary nonnegative real numbers; for $N = 0$, the formula reduces to $C = 1/\eta$. Since $\|\text{ad}(N)M\|_1 \leq 2\|N\|_1\|M\|_1$, the finite inverse (3.5) gives

$$\|\mathcal{T}F\|_r \leq C\|F\|_r, \quad \|\mathcal{P}F\|_r \leq \|F\|_r. \quad (6.4)$$

Suppose $a_0 = \|A_+\|_{r_0} < \infty$. For $0 < r \leq r_0$,

$$a := \|A_+\|_r \leq a_0(r/r_0)^\delta. \quad (6.5)$$

Choose $r > 0$ with $Ca \leq 1/8$. If $a_0 > 0$, one explicit permissible choice is

$$r = r_0 \min \left\{ 1, (8Ca_0)^{-1/\delta} \right\}. \quad (6.6)$$

The zero-input case is immediate.

On the closed ball $\|F\|_r \leq 2a$, the same map Φ as in (3.10) satisfies

$$\begin{aligned} \|\Phi(F)\|_r &\leq a + aC(2a) + C(2a)(2a) = a + 6Ca^2 \leq \frac{7}{4}a < 2a, \\ \|\Phi(F) - \Phi(G)\|_r &\leq (Ca + 4Ca)\|F - G\|_r = 5Ca\|F - G\|_r. \end{aligned}$$

Thus $q := 5Ca \leq 5/8$ is a contraction constant. The Banach fixed-point theorem yields an absolutely convergent F in this ball. Its coefficient identity is the formal fixed-point equation, so formal uniqueness identifies it with the Hahn solution constructed earlier. Consequently,

$$\|H - I\|_r \leq 2Ca \leq \frac{1}{4}, \quad \|R\|_r \leq 2a, \quad \|H^{-1}\|_r \leq \frac{1}{1 - 2Ca}, \quad (6.7)$$

where the last norm includes the constant term.

The Picard iterates provide the additional error certificate

$$\|F - F_n\|_r \leq \frac{aq^n}{1 - q}, \quad \|H - (I + \mathcal{T}F_n)\|_r \leq \frac{Caq^n}{1 - q}. \quad (6.8)$$

Indeed, the first iterate is $F_1 = A_+$ and the differences decrease by at least the factor q . This proves the analytic assertion and supplies a constructive bound rather than just a qualitative convergence theorem.

All identities now hold analytically on smaller punctured radii. In particular, (4.2) is an actual fundamental matrix on any fixed logarithmic branch there. The word “Fuchsian” here refers to the Euler form and limiting constant coefficient; we do not assume that A_+ has an ordinary Taylor expansion at zero.

6.3 Proof of necessity and small counterexamples

Assume $\eta = 0$. By Lemma 6.1, choose a positive difference $\rho = \lambda_i - \lambda_j$ and a strictly increasing sequence $\gamma_n \in \Gamma_{>0}$ with limit ρ , with no term equal to ρ . Passing to a subsequence, arrange

$$d_n := \rho - \gamma_n \leq 2^{-n}. \quad (6.9)$$

Choose a nonzero right eigenvector u of A_0 at λ_i and a nonzero left eigenvector v^T at λ_j . Normalize $U = uv^T$ so that $\|U\|_1 = 1$. Distinct eigenvalues imply $v^T u = 0$, so

$$U^2 = 0, \quad [A_0, U] = [S, U] = \rho U, \quad [N, U] = 0. \quad (6.10)$$

For any $\varepsilon > 0$ define

$$A_+(x) = \varepsilon \sum_{n \geq 1} \frac{d_n}{n} x^{\gamma_n} U. \quad (6.11)$$

The exponents remain in a bounded interval, and $\sum d_n/n \leq \sum 2^{-n}/n < \infty$. Thus the input is absolutely convergent at every radius, with norm at radius 1 tending to zero as $\varepsilon \rightarrow 0$.

The normalized formal gauge is exactly

$$H = I - \varepsilon \sum_{n \geq 1} \frac{x^{\gamma_n}}{n} U, \quad R = 0. \quad (6.12)$$

To check this, write $H = I + hU$. The nonlinear products vanish because $U^2 = 0$, and the gauge equation reduces to $(D - \rho)h = \varepsilon \sum d_n x^{\gamma_n}/n$. Its coefficient solution is $h_{\gamma_n} = -\varepsilon/n$. All these coefficients are nonresonant, so the gauge is normalized; uniqueness proves the assertion. For every $r > 0$, the factors r^{γ_n} tend to $r^\rho > 0$. Consequently $\sum r^{\gamma_n}/n = \infty$, and the normalized gauge is not absolutely convergent at any radius. This proves necessity and the smallness statement in Theorem 6.2. $\square$

6.4 Consequences and exact boundaries

Corollary 6.3 (Finite positive generators need no arithmetic condition). *If Γ is generated by finitely many positive real numbers, every absolutely convergent Γ -supported coefficient series has a convergent normalized gauge. Rational relations among the generators are irrelevant.*

Proof. Below any fixed bound, the nonnegative integer coefficient of each positive generator is bounded. Hence Γ is locally finite. No finite positive eigenvalue difference can be an accumulation point. $\square$

For example, $\langle 1, \sqrt{2} \rangle$ is locally finite even though the additive group generated by 1 and $\sqrt{2}$ is dense in $\mathbb{R}$. Replacing the positive semigroup by that group would give the wrong small-divisor diagnosis. The dangerous event is not irrationality, nor non-local-finiteness by itself: it is a semigroup accumulation specifically at a positive spectral difference.

In dimension one, there is no positive eigenvalue difference and $\eta \geq \delta$. Thus rank two is the smallest dimension for the failure mechanism. Exact resonances and near-resonances also have opposite roles. Exact resonances produce finitely many logarithmic data. An infinite sequence of near-resonances can destroy convergence while producing no additional exact resonant term at all.

Remark 6.4 (Universal quantifiers cannot be weakened silently). The counterexample uses a sequence from Γ , which need not lie in a prescribed proper generating subset E . The theorem quantifies over all inputs supported in $\Gamma_{>0}$. For one fixed input, $\eta = 0$ does not force divergence: its coefficients may decay enough, or matrix products may suppress the bad divisors. A theorem quantifying only over series supported in a smaller E is a separate problem.

7 Indirect resonance and a finite-data example

A direct inspection of the original coefficient at a resonant exponent is insufficient: earlier nonresonant terms can generate a logarithm. This is already visible in a two-dimensional system with ordinary integer powers.

Let

$$S = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}, \quad N = 0, \quad A_1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 \\ e & 0 \end{pmatrix}. \quad (7.1)$$

The only positive resonance is 2, in the lower-left entry. At exponent 1, the recurrence gives

$$H_1 = \begin{pmatrix} a & b/3 \\ -c & d \end{pmatrix}, \quad R_1 = 0.$$

At exponent 2, the lower-left entry of $A_2 + A_1 H_1$ is $e + c(a - d)$. Therefore

$$B = (e + c(a - d))E_{21}. \quad (7.2)$$

The exact no-log condition is the algebraic equation $e + c(a - d) = 0$, not merely $e = 0$. When it fails, the log-free solution subspace has dimension one, and a fundamental matrix has logarithmic degree one.

For a fully explicit example, take $a = c = 1$ and $b = d = e = 0$. The system is

$$Dy_1 = xy_1, \quad Dy_2 = 2y_2 + xy_1. \quad (7.3)$$

A fundamental matrix is

$$Y(x) = \begin{pmatrix} e^x & 0 \\ -x + x^2 \log x + \sum_{k=2}^{\infty} \frac{x^{k+1}}{(k-1)k!} & x^2 \end{pmatrix}. \quad (7.4)$$

The series is entire. Direct application of $D - 2$ to its lower-left entry gives xe^x , including the x^2 term supplied by $x^2 \log x$. The normalized gauge has $H_{22} = 1$, $H_{12} = 0$ and

$$H_{11} = e^x, \quad H_{21} = -x + \sum_{n=3}^{\infty} \frac{x^n}{(n-2)(n-1)!}.$$

In particular its resonant $x^2 E_{21}$ coefficient vanishes, as required, and $B = E_{21}$.

Now allow an arbitrary extra tail with exponents in

$$\{3 - 1/n : n \geq 2\}.$$

Every new exponent exceeds 2. Its coefficient matrices may be arbitrary Hahn data: none can appear in a word ending at the sole resonance 2. Thus the obstruction (7.2) is unchanged. The augmented semigroup has an accumulation at 3, but this is not a positive spectral difference for this system. There are only finitely many semigroup elements in a sufficiently small neighborhood of 2, so the uniform divisor gap remains positive. Absolutely convergent tails therefore also preserve analytic normalizability. This example separates support accumulation, finite logarithmic data, and analytic failure in a single calculation.

8 An explicit summability threshold at an accumulation exponent

8.1 The family and its divergent formal normalizer

Let $p > 1$ and

$$a_p(x) = \sum_{n=1}^{\infty} \frac{x^{2-1/n}}{n^p}, \quad S = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad U = E_{12}. \quad (8.1)$$

Consider

$$DY = (S + a_p(x)U)Y. \quad (8.2)$$

The input series is absolutely convergent at every positive radius: its exponents belong to $[1, 2]$ and $\sum n^{-p}$ converges. It is holomorphic on the logarithmic cover and tends to zero as $x \rightarrow 0^+$. Its well-ordered support approaches the positive spectral difference 2 from below.

Proposition 8.1 (Exact threshold). *The normalized formal gauge and resonant matrix are*

$$\hat{H}_p = I + \hat{h}_p U, \quad \hat{h}_p = - \sum_{n=1}^{\infty} n^{1-p} x^{2-1/n}, \quad R = 0, \quad (8.3)$$

and $B = 0$. The gauge is absolutely convergent at some positive radius if and only if $p > 2$; in that case it converges at every positive radius.

Proof. Since $[S, U] = 2U$ and $U^2 = 0$, the coefficient equation is $(D - 2)\hat{h}_p = a_p$. Dividing the coefficient at $2 - 1/n$ by $-1/n$ gives $-n^{1-p}$. There are no product terms and no input at the resonance itself, so $R = 0$. At any fixed positive radius, $r^{2-1/n}$ tends to r^2 , so absolute convergence is equivalent to that of $\sum n^{1-p}$. $\square$

Although the generated semigroup contains $2 = 1 + 1$, the exact resonant coefficient remains zero: the only possible matrix product is a multiple of $U^2 = 0$. This illustrates why membership of a resonance in the semigroup alone is not the obstruction certificate.

8.2 Renormalized analytic solutions for every $p > 1$

The formal series can diverge while the actual differential equation still has a particularly explicit solution. For $t = -\log x$ define

$$\Psi_p(t) = \sum_{n=1}^{\infty} n^{1-p} (e^{t/n} - 1), \quad (8.4)$$

$$h_p(x) = -x^2 \Psi_p(-\log x). \quad (8.5)$$

The subtraction is made *before* summation.

Theorem 8.2 (Analytic realization by renormalization). *For every $p > 1$, Ψ_p is entire and*

$$\Psi_p(t) = \sum_{k=1}^{\infty} \zeta(p+k-1) \frac{t^k}{k!}. \quad (8.6)$$

The function h_p is holomorphic on the logarithmic cover, satisfies $(D-2)h_p = a_p$ and $h_p(1) = 0$, and gives the fundamental matrix

$$Y_p = (I + h_p U)x^S. \quad (8.7)$$

For $p > 2$, it agrees with the convergent formal normalizer up to the homogeneous term

$$h_p = \widehat{h}_p + \zeta(p-1)x^2. \quad (8.8)$$

For $1 < p \leq 2$, the defining difference series still converges, but its two pieces cannot be separately summed absolutely.

Proof. For $|t| \leq T$,

$$n^{1-p}|e^{t/n} - 1| \leq Te^T n^{-p}.$$

The series (8.4) therefore converges uniformly on compact sets, proving that Ψ_p is entire. Expanding each exponential and using absolute convergence gives

$$\sum_{n \geq 1} \sum_{k \geq 1} \frac{|t|^k}{k! n^{p+k-1}} \leq \zeta(p)(e^{|t|} - 1) < \infty.$$

Interchanging the sums proves (8.6).

Since $D = -d/dt$ on functions of t and $x^2 = e^{-2t}$,

$$(D - 2)(-x^2 \Psi_p(t)) = x^2 \Psi'_p(t) = x^2 \sum_{n \geq 1} n^{-p} e^{t/n} = a_p(x).$$

The value at $x = 1$ is zero. This scalar equation and $U^2 = 0$ verify (8.7) directly. If $p > 2$, one can split (8.4) into two absolutely convergent sums; their difference is precisely (8.8). If $p \leq 2$, the series of subtracted constants is $\sum n^{1-p}$ and diverges, so that split is not legitimate. $\square$

8.3 Every finite asymptotic prefix is nevertheless correct

For $N \geq 1$, set

$$h_{p,N}(x) = - \sum_{n=1}^N n^{1-p}(x^{2-1/n} - x^2), \quad S_{p,N}(t) = \sum_{n > N} n^{1-p}(e^{t/n} - 1). \quad (8.9)$$

For $0 < x < 1$, $t = -\log x > 0$ and

$$h_p - h_{p,N} = -x^2 S_{p,N}(t). \quad (8.10)$$

Lemma 8.3 (Elementary two-sided tail certificate). *For $p > 1$, $N \geq 1$, $t > 0$,*

$$\frac{t}{p-1}(N+1)^{1-p} \leq S_{p,N}(t) \leq \frac{te^{t/(N+1)}}{p-1} N^{1-p}. \quad (8.11)$$

Proof. For $u \geq 0$, $u \leq e^u - 1 \leq ue^u$. Thus

$$t \sum_{n > N} n^{-p} \leq S_{p,N}(t) \leq te^{t/(N+1)} \sum_{n > N} n^{-p}.$$

The integral test bounds the final sum between $(N+1)^{1-p}/(p-1)$ and $N^{1-p}/(p-1)$. $\square$

Proposition 8.4 (Asymptotic realization without absolute summation). *For every fixed $N \geq 1$ and every $p > 1$,*

$$h_p(x) + \sum_{n=1}^N n^{1-p} x^{2-1/n} \sim -(N+1)^{1-p} x^{2-1/(N+1)} \quad (x \rightarrow 0^+). \quad (8.12)$$

In particular, h_p realizes all finite Poincaré prefixes of $\widehat{h}_p$, including in the divergent range $1 < p \leq 2$. Adding any constant multiple of x^2 leaves every such prefix unchanged.

Proof. For fixed N , isolate the first term of the tail:

$$S_{p,N}(t) = (N+1)^{1-p}(e^{t/(N+1)} - 1) + S_{p,N+1}(t).$$

The upper estimate in Lemma 8.3 shows that the last term is $O(te^{t/(N+2)})$, which is little-oh of $e^{t/(N+1)}$. Consequently $S_{p,N}(t) \sim (N+1)^{1-p}e^{t/(N+1)}$. The left side of (8.12) equals $x^2 \sum_{n \leq N} n^{1-p} - x^2 S_{p,N}(t)$. The finite constant sum is negligible compared with $e^{t/(N+1)}$, proving the formula. Finally, $x^2 = o(x^{2-1/n})$ for every fixed n . $\square$

This is a precise boundary between three assertions often conflated in transseries manipulations. The series is formally valid. It is asymptotically realized to each fixed finite order. It is absolutely summable only for $p > 2$. Neither of the first two assertions implies the third. The undetectable Cx^2 term is also the natural analogue, for this accumulating scale, of the repository's flat-function ambiguity.

9 The truncation crossover and an explicit uniform remainder

Fixed N describes successive asymptotic prefixes; $N \rightarrow \infty$ at fixed x describes convergence of the renormalized sum. The interesting transition occurs when both grow and N is proportional to $t = \log(1/x)$. Set

$$f_p(s) = s^{1-p}(e^{1/s} - 1), \quad \Phi_p(c) = \int_c^\infty f_p(s) ds = \int_0^{1/c} u^{p-3}(e^u - 1) du. \quad (9.1)$$

Both integrals converge for $p > 1$ and $c > 0$.

Theorem 9.1 (Certified accumulation-scale crossover). *Let $p > 1$, $t > 0$, $N \geq 1$, and $c = N/t$. Then*

$$S_{p,N}(t) = t^{2-p} \left(\Phi_p(c) - \frac{f_p(c)}{2t} - \frac{f'_p(c)}{12t^2} + \mathcal{E}_{p,N,t} \right), \quad (9.2)$$

with the explicit bound

$$|\mathcal{E}_{p,N,t}| \leq \frac{|f'''_p(c)|}{360t^4}. \quad (9.3)$$

In particular, the expansion is uniform when c ranges over a compact subinterval of $(0, \infty)$, with p fixed. If $c > 0$ is fixed and $t \rightarrow \infty$ along values with $ct \in \mathbb{N}$, then

$$t^4 \mathcal{E}_{p,ct,t} \longrightarrow \frac{f'''_p(c)}{720}. \quad (9.4)$$

Proof. From the definition,

$$S_{p,N}(t) = t^{1-p} \sum_{n > N} f_p(n/t).$$

Thus $t^{p-2}S_{p,N}(t)$ is a right-endpoint Riemann sum with mesh $h = 1/t$ beginning strictly after $c = Nh$.

We record the needed half-line Euler–Maclaurin formula, including the constants. For a smooth integrable function f whose relevant derivatives are integrable and decay at infinity,

$$h \sum_{k=1}^{\infty} f(c+kh) = \int_c^\infty f(s) ds - \frac{h}{2}f(c) - \frac{h^2}{12}f'(c) + \frac{h^4}{720}f'''(c) + \mathcal{R}_4, \quad (9.5)$$

$$|\mathcal{R}_4| \leq \frac{h^4}{720} \int_c^\infty |f^{(4)}(s)| ds. \quad (9.6)$$

This follows by integrating by parts with the periodic Bernoulli polynomials on successive mesh intervals. The coefficients are $B_2/2! = 1/12$ and $B_4/4! = -1/720$; the boundary terms at infinity vanish. The periodic fourth Bernoulli polynomial divided by $4!$ has absolute value at most $1/720$, giving (9.6). This also proves the version on an infinite interval by first using a finite number of mesh intervals and passing to the limit.

For our function,

$$f_p(s) = \sum_{k=1}^{\infty} \frac{s^{-(p+k-1)}}{k!}. \quad (9.7)$$

The series and each derivative converge uniformly on $[c, \infty)$ after the appropriate integrable power bound. Every summand is completely monotone, so $f_p^{(4)} \geq 0$ and $f_p''' \leq 0$. Its derivatives tend to zero at infinity; hence

$$\int_c^{\infty} |f_p^{(4)}(s)| ds = -f_p'''(c) = |f_p'''(c)|.$$

Combining the explicit fourth-order term in (9.5) with its remainder gives the factor $2/720 = 1/360$ in (9.3). Substituting $h = 1/t$ proves (9.2).

For the limit, apply the same formula two derivatives further. The sixth-derivative remainder is $O(h^6)$ for fixed c , since the positive inverse-power expansion gives integrability of every derivative from the first onward. Thus the fourth-order coefficient is $f_p'''(c)/720$ and the unrecorded terms are $O(h^6)$, proving (9.4). Bounds on these derivatives are uniform on a compact c -interval, which also proves the stated uniformity. $\square$

9.1 Reading the three regimes

If $N/t \rightarrow \infty$ and $N \rightarrow \infty$, the elementary bounds already imply

$$S_{p,N}(t) \sim \frac{tN^{1-p}}{p-1}. \quad (9.8)$$

Indeed, the ratio of the upper and lower bounds tends to one. When $N/t \rightarrow c \in (0, \infty)$, the leading term is instead $t^{2-p}\Phi_p(c)$. The exponent $p = 2$ is critical: after factoring out x^2 in the solution error (8.10), the transition scale has no extra power of t at $p = 2$, grows for $1 < p < 2$, and decays for $p > 2$.

At the critical exponent,

$$\Phi_2(c) = \int_0^{1/c} \frac{e^u - 1}{u} du = \sum_{k=1}^{\infty} \frac{c^{-k}}{k k!}. \quad (9.9)$$

For general $p > 1$, the corresponding absolutely convergent series is

$$\Phi_p(c) = \sum_{k=1}^{\infty} \frac{c^{-(p+k-2)}}{k! (p+k-2)}. \quad (9.10)$$

These formulas also supply stable numerical evaluations of the transition function.

The crossover is a joint truncation statement. It does not append a new canonical term at exponent 2 to the formal series, nor does it justify exchanging an infinite Hahn sum with analytic evaluation. The dependence on how N grows with t is exactly the information that a fixed finite prefix leaves unresolved.

9.2 Numerical illustration

Table 1 uses $p = 2$, $c = 1$, so $N = t$. Write

$$Q(t) = \Phi_2(1) - \frac{f_2(1)}{2t} - \frac{f_2'(1)}{12t^2}.$$

The error column is computed from the convergent Hurwitz-zeta expansion

$$S_{p,N}(t) = \sum_{k=1}^{\infty} \frac{t^k}{k!} \zeta(p+k-1, N+1), \quad (9.11)$$

which follows by absolute summation exactly as in (8.6). The bound is a theorem, while the displayed decimals are high-precision floating-point illustrations, not interval certificates.

$t = N$	$S_{2,N}(t)$	$S_{2,N}(t) - Q(t)$	Proved error bound
20	1.275868641749	-7.48089×10^{-7}	1.50037×10^{-6}
40	1.296654652769	-4.68538×10^{-8}	9.37734×10^{-8}
80	1.307220654853	-2.92990×10^{-9}	5.86084×10^{-9}
160	1.312546962496	-1.83143×10^{-10}	3.66302×10^{-10}

Table 1: The critical accumulation crossover. Decimal computations use 85-digit working precision; the bound is (9.3).

10 Computation, verification, and a route to Lean

10.1 Finite exact normalizer

The supplied `verification/verify.py` represents a matrix series as a dictionary from rational exponents to exact SymPy matrices. For a finite positive rational support and a rational cutoff, it enumerates the generated semigroup below the cutoff, processes exponents in increasing order, and uses (3.5) and (3.9). After constructing H and R , it checks the complete coefficient residual of (3.7), including all matrix products, at every enumerated exponent. It also checks the normalization masks and nilpotence of B .

The code accepts commuting nilpotent N rather than assuming a semisimple constant matrix. This matters: the finite geometric inverse and its noncommutative products are among the places most susceptible to a sign or projection error. Exact checks include the symbolic polynomial $e + c(a - d)$, sharp chains of ranks 1 through 7, and examples with a nontrivial repeated-eigenvalue Jordan block.

The recorded run, with seed 20260929, has the following scope:

Verification item	Count
Exact finite gauge systems	33
Exact coefficient checks, including scalar counterexample checks	292
Crossover parameter cases	36

The numerical cases use $p \in \{1.5, 2, 2.5\}$, $c \in \{0.5, 1, 2\}$, and $t \in \{20, 40, 80, 160\}$. They compare (9.11) with both (8.11) and (9.3), and record convergence toward (9.4). The environment was Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0.

These are reproducible finite consistency checks, not substitutes for the proofs. They do not establish the infinite-support recursion, the universal quantifiers in Theorem 6.2, or arbitrary-support well-ordering. No Lean code in this package has been claimed to prove those statements.

10.2 A formalization decomposition

A direct Lean development can be organized around four interfaces, each with a narrow mathematical contract.

Support arithmetic. Reuse the repository’s two-factor additive Neumann bridge to prove the positive-semigroup finite-word lemma. The additional content is the real lower bound δ and the conversion of a fixed exponent bound into a natural-number word-length bound. Do not assume that bounded supports are finite.

Finite-dimensional homological algebra. Formalize the spectral-difference masks, their commutation with $\text{ad } N$, the identity $(\text{ad } N)^{2m-1} = 0$, and the finite inverse (3.5). This component is independent of Hahn series and can be checked over a characteristic-zero field with suitable eigenvalue data.

Strong Hahn derivation and recursive gauge. Lift those operators coefficientwise and prove the valuation improvement (3.11). The derivative’s product rule, the normal-form equation, and constants in $\mathcal{K}[L]$ should be explicit interfaces, not informal consequences of a field name. The repository’s block-antiderivative API is related algebra, but it is not already a formalization of this matrix construction.

Analytic realization. Build the weighted ℓ^1 convolution algebra, prove the operator norm bounds and contraction estimates, and only then supply a branchwise analytic evaluation map. The universal converse can be formalized separately with the rank-one family. This prevents analytic summation from being silently built into the formal Hahn datatype.

The suggested theorem targets are therefore finite-word support, normalized gauge existence and uniqueness, nilpotent log filtration, finite ancestry, absolute-convergence preservation, and its sharp failure criterion. None is asserted to be present in the inspected repository merely because related names already exist there.

11 Further research questions and concrete next targets

11.1 Universal convergence for a fixed smaller input support

Theorem 6.2 quantifies over all inputs supported in Γ . Fix instead a generating set $E \subsetneq \Gamma$ and allow coefficients only on E . Is the absence of a left-accumulating spectral difference in Γ still necessary? A bad exponent might be reachable only through a long product whose coefficients or matrix directions are constrained. A useful first target is a triangular three-dimensional system with two permitted matrix units and an accumulating two-letter sumset. This asks whether an additive obstruction must have a realizable noncommutative path.

11.2 Weighted convergence in the zero-gap case

For the square-zero family, convergence is exactly a weighted summability condition after division by $\gamma - \rho$. For general matrices, products of several small divisors appear. Construct a weight on exponent paths that is both sufficient for a Banach fixed-point argument and close to necessary. One candidate is the least submultiplicative majorant of the homological inverse norms. The main issue is whether cancellations permit substantially larger classes than any absolute path majorant can capture.

Editorial note (ProveIt, 2026-09-29). The two preceding questions (a fixed smaller input support; weighted convergence in the zero-gap case) are answered, for finite acyclic patterns with a diagonal real constant term, a well-ordered support on each edge and independently variable edge coefficients, by the later package `Analysis/Transseries/docs/series-and-transseries/Path_Sensitive_Small_Divisors_Hahn_Fuchsian/path_sensitive_hahn.tex` (batch 52; not reviewed). Universal convergence holds exactly when no endpoint spectral difference of a directed path is a left accumulation point of that path’s own exponent sumset (its `thm:gap`), so an accumulating difference in Γ that no path realizes is harmless; for independent weighted inputs it holds exactly when every path-kernel norm is finite, and that norm is the exact multilinear operator norm of the path contribution (its `thm:weighted`). Correlated inputs can do better: its example in Section `sec:cancellation` cancels two accumulating paths. Its rank- r resonant chain converges exactly when $\sum n^{r-1} |c_n| < \infty$ (its `thm:chain`), which at $r = 2$ is the threshold $p > 2$ of Proposition 8.1. For the complete triangular pattern with the whole semigroup on every edge, its `thm:gap` reduces to the triangular case of Theorem 6.2 with a diagonal constant term, consistently; Theorem 6.2 also allows nontriangular perturbations and a nilpotent part, which that package excludes. Nonlinear counterparts of Theorem 6.2 are proved in `Nonlinear_Hahn_Dulac_Finite_Resonance_Control/` (its `thm:universal`) and `Nonlinear_Hahn_Fuchsian_Algebraic_Convergence_Loci/` (its `thm:gap`), in the same directory.

11.3 Renormalization compatible with multiplication and gauge change

The difference sum in (8.4) gives one natural analytic realization of a divergent normalizer, selected by a boundary value at $x = 1$. Can this be extended to a summation operation on an algebra of accumulating-support series that commutes with D and respects products? Different normalizing conditions add resonant homogeneous terms. A precise first task is to classify all such realizations for upper-triangular rank-three systems and identify the additional compatibility identities caused by nonzero products of matrix units.

11.4 Uniform spectral parameters and moving accumulation resonances

Let the positive eigenvalue difference vary through an accumulation point of Γ . The constant C in (6.3) blows up, even though individual solutions may continue after renormalization. Determine a uniform scaling limit with $\rho - \gamma_n$ and the spectral displacement competing. The model $\gamma_n = 2 - 1/n$ suggests transition parameters involving both $N/\log(1/x)$ and the displacement times N . A uniform theorem should separate exact resonance, near-resonance, and the accumulation boundary.

11.5 Iterated-logarithmic coefficient fields

Replace the constant matrices A_α by controlled Hahn series in $\log(1/x)$ and further iterated logarithms. Which parts of the finite matrix certificate survive, and when does one additional logarithmic level suffice? This connects directly with the earlier residue-depth results in ProveIt. A first tractable case keeps the outer exponent gap positive and allows one rational function of $\log(1/x)$ in each coefficient. The scalar block inverse then interacts with the matrix spectral inverse.

11.6 Higher-rank exponent groups and the loss of finite word depth

In a non-Archimedean exponent group, a positive generator can be smaller than every positive real multiple of another scale. The bound $k\delta \leq r_*$ need not bound k . Find an intrinsic replacement for the finite-ancestry hypothesis, perhaps a no-infinite-resonant-path condition. The intended conclusion is not necessarily a finite ordinary matrix but a controlled transfinite filtration. A counterexample to a naive finite-depth extension would be as informative as a sufficient theorem.

11.7 Effective critical ancestry and exact real presentations

For finite rational generators, the included algorithm is elementary. For finite algebraic generators, exact comparison and equality are still available but implementation costs differ. For infinite computable supports, identify representation hypotheses under which E_{crit} can actually be extracted. A useful deliverable would be a theorem separating finite dependence, semidecidable resonance detection, and decidable complete certificates, rather than conflating these three levels.

11.8 Geometry of the logarithmic strata

The loci $B^{d+1} = 0$ are algebraic in critical coefficient data. Determine their irreducible components and generic codimensions for a prescribed spectral graph. Even a three- or four-level chain can have cancellation between several paths. The polynomial $e + c(a - d)$ is the first nontrivial example. Higher-rank cases could give a useful classification of when logarithms disappear under parameter tuning and when a small perturbation necessarily restores them.

11.9 Irregular systems and exponential transseries sectors

For an irregular system, exponentially small or large normal-form sectors introduce a second source of divergence, distinct from accumulation at a finite spectral difference. Develop a model in which one irregular exponential sector and one accumulating real-exponent support coexist. The immediate goal is a theorem separating a Borel-summation obstruction from the absolute small-divisor obstruction proved here. Existing resurgence terminology alone does not identify which mechanism is active.

11.10 Formal verification of the sharp equivalence

The most valuable initial proof-assistant target is not the numerical crossover but Theorem 6.2: its quantifiers expose the formal/analytic distinction particularly clearly. A staged project can first verify the square-zero counterexample and the finite-dimensional inverse, then the ℓ^1

contraction, and finally the positive-Hahn support layer. The finite-ancestry certificate would make a natural exact computational interface on top of that verified core.

12 Conclusion

The formal normal form and the analytic convergence problem have different answers. Positive well-ordered supports suffice for a unique normalized Hahn gauge. Exact resonances reduce to a finite nilpotent matrix, and even an infinitely populated low-exponent region contributes only finitely many ancestors to its logarithmic obstruction. But absolute convergence depends on an additional relation between the exponent semigroup and the spectrum. The required relation is exactly a uniform gap from the nonresonant spectral differences.

The explicit family with exponents $2 - 1/n$ demonstrates all the distinctions at once. Its coefficients converge for $p > 1$, its formal normalizer converges only for $p > 2$, and its renormalized analytic solution realizes every finite asymptotic prefix throughout the whole range $p > 1$. The transition $N \asymp \log(1/x)$ quantifies the information lost when an accumulating formal expansion is treated as an ordinary convergent sum.

These results offer a specific extension of the repository’s support, resonance, and flatness calculus. They also delimit that extension: the proofs concern the stated real-exponent Fuchsian class, not general transseries equations; the certificate is mathematically finite but not automatically computable from arbitrary infinite presentations; and historical priority remains a separate question from the correctness of the supplied proofs.

A Proof and provenance audit

A.1 Assumptions that must not be dropped

The formal construction uses positive real input exponents with a common positive least element, well ordering, finite matrix dimension, and a real-spectrum constant matrix. The log-degree classification fixes the ambient algebra $\mathcal{K}[L]$. The analytic criterion uses absolute ℓ^1 convergence in a fixed Jordan basis and quantifies over the whole positive semigroup. The uniform crossover holds for $p > 1$ and for $N \geq 1$; its compact-parameter assertion keeps N/t away from zero and infinity.

Each restriction has a role. The positive lower bound controls the number of input factors. Well ordering controls finite coefficient convolution. Real eigenvalues allow real monomials x^{λ_i} in the chosen Hahn field. Absolute norms justify products and analytic evaluation. The whole-semigroup quantifier is needed for the converse construction. The fixed logarithmic algebra makes minimal log degree a well-posed invariant.

A.2 Dependency map

Lemma 2.1 and Lemma 3.1 imply the formal normalization. The normal form, the constant-field lemma, and the nilpotence of B imply the log filtration. Finite-word dependence gives the ancestry certificate. The analytic forward direction adds a uniform operator bound and Banach contraction; the converse uses only the spectral gap characterization and a rank-one eigenmatrix. The model and crossover are proved independently by scalar summation and Euler–Maclaurin, so they do not depend on unproved analytic realization of arbitrary Hahn series.

A.3 Repository inspection boundary

The snapshot was pinned to 04473354a0f3edff2d6c365caf26170ca6b88735. The group index, canonical-package status documentation, the two-factor Hahn-support Lean module, the block-antiderivative module, and the earlier residue-obstruction package README were inspected through the connected GitHub tools. The canonical large TeX volume exceeded the connector’s content limit; no claim of an exhaustive line-by-line audit of that volume or the whole repository is made. The supplied provenance JSON records file paths, blob identifiers, and the scope of inspected ranges. No repository file was modified.

A.4 Novelty and verification boundary

The normal-form architecture specializes to classical Fuchsian theory, and the finite-obstruction philosophy has an explicit repository antecedent. The spectral accumulation equivalence, its sharp analytic examples, the finite-ancestry formulation, and the uniform crossover are the proposed research additions developed here. The literature check was targeted, not exhaustive; no independent peer review has occurred. The exact scripts check finite algebraic instances, while the numerical results illustrate bounds proved in the text. Neither is represented as a machine proof of the general theorems.

References

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- [3] Vladimir Reshetnikov, ProveIt contributors, *TransseriesBlockAntiderivative.lean*, same repository snapshot. <https://github.com/VladimirReshetnikov/ProveIt/blob/04473354a0f3edff2d6c365caf26170ca6b88735/Analysis/Transseries/Lean/Transseries/TransseriesBlockAntiderivative.lean>.
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- [5] Shira Tanny and Sergei Yakovenko, *On local Weyl equivalence of higher order Fuchsian equations*, arXiv:1412.7830 (2014). <https://arxiv.org/abs/1412.7830>. The spelling of “Fuchsian” follows the arXiv title; p. 6 of the PDF records the classical resonant polynomial normal form.

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